Separating Neural Scale Response and Score Memory in Online Interval Calibration

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Working draft for student review, 18 October 2026

Abstract

Normalized-score calibration combines a neural forecaster's current scale with historical forecast errors divided by their originally issued scales. We study these two pathways separately in paired volatility-pulse simulations, keeping trained weights fixed and using the same observations across configurations. An initial five-seed GRU/TCN comparison is sensitive to training randomness. We replicate with three new training datasets and five paired seeds per dataset, retaining both current-scale and archived-score decomposition orders. The entry architecture contrast keeps its direction across new datasets, but training-randomness sensitivity still includes zero; the recovery contrast does not replicate reliably. In the reference Gaussian entry comparison, TCN has greater current-channel seed variability than GRU, while archive variability can dominate within a dataset. A post-hoc common-rank-level check retains the negative mean Gaussian entry contrast while revealing sensitivity in smaller effects. A separate public-FX extension compares absolute interval quality and descriptive pathway differences. These findings do not establish a general architecture ranking or a coverage guarantee; all fitted models and failed replication outcomes remain.

1 Research question and contribution scope

Financial uncertainty intervals should expand when risk rises and contract when a temporary disturbance ends. A learned scale model and an online calibrator can each retain information about that disturbance. Their combination may behave differently from either component alone. We ask: *where does variability across fitted neural models enter normalized-score calibration at shock entry and recovery?*

This controlled deep-learning study learns scales from raw past returns; a zero center isolates uncertainty adaptation. It examines a computational pathway without establishing trading gains or deployable conditional coverage.

Underspecification already explains why strong in-domain validation need not constrain deployment behavior [1]; HARNet studies initialization stability in neural volatility prediction [2]. Neural conformal control [3], jointly adapting scores and thresholds [4], retrospective residual updates [5], strongly adaptive experts [6], and scorecasting with feedback [7] are established. We do not claim these ideas as new. AdaptNC updates score parameters with replay; RetroAdj recomputes leave-one-out residuals. Our experiment fixes neural weights and separately varies current and archive contexts.

The intended increment is an explicit separation of *current neural scale* and *historically issued score denominators*, with paired observations, two decomposition orders, and independent training-data repeats. The initial crossed analysis was exploratory; its specific replication questions were fixed before new training and test generation. Evidence that fails to replicate must narrow the conclusion.

2 Crossed pathways and evaluation

For a fixed trained model, let $s_t^a > 0$ be the scale predicted from the last $a \in \{16, 64\}$ observations, excluding y_t. Define

$$q_t^{b,W} = \mathrm{OrderStat}_k \left\{ \frac{|y_i|}{s_i^b} : i = t-W, \ldots, t-1 \right\}, \quad h_t^{a,b,W} = s_t^a q_t^{b,W}, \quad k = \lceil (W+1)(1-\alpha) \rceil. \tag{1}$$

Here $b \in \{16, 64\}$, $W \in \{32, 128\}$ and $\alpha = .1$. The deployed diagonal has $a = b$; the full eight-cell factorial changes the current and archive contexts independently. Warmup uses the last 128 scores of a separate validation path, with predictor-specific issued denominators.

For each cell and phase, $E(a, b, W)$ is its event-minus-matched-control mean interval score. Write $D(a, b) = E(a, b, 128) - E(a, b, 32)$. The diagonal context-by-window interaction is $J = D(64, 64) - D(16, 16)$, and

$$J = \underbrace{D(64,64) - D(16,64)}_{C_{b64}} + \underbrace{D(16,64) - D(16,16)}_{A_{a16}}, \tag{2}$$

$$= \underbrace{D(64,16) - D(16,16)}_{C_{b16}} + \underbrace{D(64,64) - D(64,16)}_{A_{a64}}. \tag{3}$$

These are exact computational differences, not unique causal mediation fractions. Correlated component interval endpoints cannot be added. C and A are channel-by-window interaction components, not channel main effects.

Training and tests. The GRU [9] has 2041 parameters; the causal dilated TCN [10] has 1897 and receptive field 64. Both use features $(y/3, \log(1 + |y|))$, Gaussian scale loss, two contexts, data and paired sampler prefixes. Each target resets state. Adam and validation early stopping are fixed; total optimization steps may differ between architectures. Context length is not effective learned memory. Training uses 256 paths of length 1024; validation uses 64 paths and 8192 shared sampled targets. Each of three new datasets trains five seed pairs. The original five seed indices are reused: independence refers to new training/validation and test data, not 15 independent initialization/sampling draws.

The initial frozen grid has 30 cases: Gaussian, unit-variance t5 and stochastic volatility, five pulse durations and two amplitudes. The new replication fixes Gaussian/t5, amplitude 3 and durations 32/128, with 128 independent paired paths per family. Pulses begin at 384; recovery64 follows their exit. Families have separate noise streams; durations share their family stream, so both duration conditions have identical entry32 and are not independent replications. Event labels and true scales enter generation and evaluation only, never the model or quantile.

Scores and uncertainty. We report interval score $2h + (2/\alpha)(|y| - h)_+$ [8], coverage, width, oracle coverage deficit and excess width. The two reference contrasts are GRU J minus TCN J for Gaussian D128/A3, at entry and recovery64. The 4000-draw paired-path bootstrap averages all 15 fixed fitted pairs before resampling; 97.5% marginal intervals provide a two-phase Bonferroni approximation. Crossed dataset/seed/path resampling is sensitivity only: shared seed draws stay shared across datasets, with only three datasets and five seeds. Secondary intervals are descriptive. Finite quantile ranks also differ: ideal continuous exchangeable-score coverage is 30/33 for W32 and 117/129 for W128. The contrast therefore includes rank discretization as well as history length; this benchmark is not a coverage guarantee for our dependent scores.

3 Results and interpretation

Initial study motivated replication. In the initial five-seed Gaussian D128/A3 study, GRU-minus-TCN J is -0.05287 at entry and $+0.03276$ at recovery64. Fixed-fit 97.5% intervals exclude zero, but crossed seed/path intervals include zero. The entry seed range is $[-0.42866, +0.47620]$; omitting one seed can reverse the mean.

Independent training-data replication. With 15 new paired fits, the reference contrasts are -0.1986 at entry, fixed-fit 97.5% interval $[-0.2494, -0.1485]$, and $+0.0069$ at recovery64, interval $[-0.0052, +0.0192]$. Crossed sensitivity ranges are $[-0.4255, +0.0179]$ and $[-0.0589, +0.0673]$. All initialization outcomes remain in the analysis; these are interactions, not absolute interval-score rankings. Entry has the same negative mean direction in all three datasets, but the crossed interval includes zero. Recovery changes direction across datasets; omitting panel 3 reverses its mean. The earlier positive recovery contrast therefore did not replicate reliably.

Phase	Panel 1	Panel 2	Panel 3
Entry	-0.3101	-0.2547	-0.0309
Recovery64	-0.0110	-0.0107	+0.0425

Entries average five paired seeds per new training-data panel. Shared seeds and test paths preclude six independent significance tests.

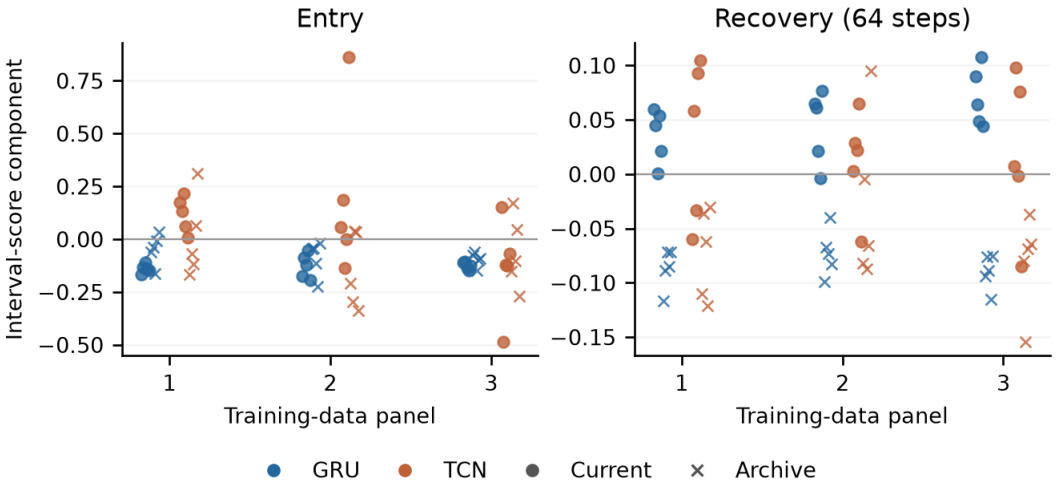


Figure 1: All five seeds per architecture and new dataset, Gaussian D128/A3. Current circles denote C_{b64} and archive crosses A_{a16}. Point spread describes training variability, not causal variance attribution.

TCN entry current seed standard deviation exceeds GRU's in all six panel-by-route comparisons, but exceeds its own archive spread in only $4/6$. Panel 1 has larger archive spread in both routes. These correlated comparisons are descriptive; Appendix A retains the component means and absolute interval costs.

Interpretation requires the score as well as the response. A small total can conceal opposing components. At recovery, GRU's $C_{b64} = +0.0504$ and $A_{a16} = -0.0830$ partly cancel; TCN's are $+0.0210$ and -0.0606. Moreover, the mixed derivative of expected Gaussian interval score for $h = sq$ changes sign (Appendix A). Thus signed J combines scale response, archive quantiles and score curvature; it cannot label memory as beneficial or harmful. Neither the algebraic decomposition nor the finite input-dependency bound is presented as a novel theorem.

4 Sensitivity, financial extension, and boundaries

Finite-rank sensitivity on the same saved paths. We freeze a post-hoc check using all 30 Stage3 models and the already-viewed Gaussian/t5 D128/A3 paths. A randomized rule uses ranks 29/30 with probabilities .30/.70 at W32, and 116/117 with .90/.10 at W128. Both have ideal exchangeable-score rank level .90. We integrate each candidate interval's *metric*, not its width before scoring. This lowers both original ideal rank levels. Within either rule, comparing W32 with W128 still changes score count and history span. This is rule sensitivity, not causal removal of a discretization effect.

Phase	Original rank	Randomized rank	Change
Entry	-0.1986	-0.1946	+0.0040
Recovery64	+0.0069	+0.0043	-0.0027

Entries are GRU-minus-TCN J for Gaussian; the change uses paired rules on common bootstrap draws. Fixed-fit 97.5% change intervals are $[-0.0124, +0.0210]$ at entry and $[-0.0103, +0.0048]$ at recovery. Crossed sensitivity intervals still contain zero; recovery panel directions remain mixed. TCN's own near-zero entry mean changes sign. For t5 recovery, the GRU-minus-TCN contrast changes by $+0.00936$ with fixed-fit 95% interval $[+0.00230, +0.01661]$, but crossed $[-0.00322, +0.02139]$. Rank choice therefore matters for some smaller quantities. This check cannot repair the failed independent recovery replication.

Public FX: absolute quality and a descriptive contrast. We use the same frozen ECB [12] USD/JPY/GBP/CHF-per-EUR archive through 9 October 2026. Training and per-asset standardization use data through 2018; checkpoint selection and calibrator-scale preparation use 2019. Each currency replays its own 2020 history before scoring the already-viewed 2021–2026 period. Ten new pooled GRU/TCN fits retain all five paired seeds. Training-only constant, EWMA16/64 and GARCH scales provide classical controls. Native, rolling32/128, ACI [11], author SF-OGD and SAOCP comparisons are retained (Appendix B); no test-based winner or event subset is selected.

	Native		Rolling128	
Model	Coverage	IS	Coverage	IS
GRU	90.05%	3.178	90.70%	3.219
TCN	91.01%	3.166	90.68%	3.204
GARCH	92.78%	3.240	90.45%	3.192

Context64 neural results average seeds, then currencies; IS is in training-standard-deviation units. Columns were chosen before the new run; Appendix B includes all six calibrators. Both neural models have lower native mean IS than GARCH. Under every online calibrator, their pooled mean IS is higher than GARCH's; uncertainty varies by comparison. This reversal motivates evaluating predictor–calibrator combinations rather than declaring a general neural advantage.

For FX, $D_{\mathrm{FX}}(a, b)$ is the observed mean score difference W128-minus-W32, and $J_{\mathrm{FX}} = D_{\mathrm{FX}}(64, 64) - D_{\mathrm{FX}}(16, 16)$, with both decomposition orders retained. There is *no matched control or known shock/recovery* on this history. Joint date-block intervals preserve shared currency dates and are conditional on the saved fits; four correlated currencies and five seeds are not independent market histories.

Scope. Evidence is specific to small zero-center neural scale models, a fixed training procedure, the simulated generator and one viewed FX archive. The study supports a reproducible separation of computational pathways and identifies where apparent contrasts fail to generalize across fits or settings. It establishes neither deployable coverage nor trading gains. Training-randomness uncertainty and the recovery failure remain central, regardless of additional sensitivity checks.

References

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Reproducibility and draft status

Frozen protocols, checkpoints, complete metrics and source/input hashes are available at https://github.com/Lebmar/DL-project/tree/codex/stage2-memory-study. Writers refuse silent overwriting. Stage3 statistics remain unchanged; new sensitivity and FX displays are read from their separately saved outputs. Main text ends on page 4; references and appendices are extra. AI tools assisted implementation, checking and drafting with course permission. This is a working report for the student's interpretation and revision; it has not been submitted by the assistant.

A Synthetic design and interpretation details

Fixed training procedure. All synthetic models use GRU hidden size 24 or TCN 12 channels with kernels $(4, 3, 3, 3, 3)$ and dilations $(1, 2, 4, 8, 16)$, clipped log-scale $[-3, 3]$, Adam learning rate .002, gradient norm clipping at 5, batch size 128 and 32 balanced-context batches per epoch. Validation chooses the absolute minimum NLL; patience 8 uses tolerance 10^{-4} after at least 20 epochs, with maximum 60. Shared sampler prefixes do not imply equal stopping times. The three Stage3 training seeds are 314259/314359/314459, with separate validation seeds and target-pair seeds; each panel reuses fit seeds 4101–4105. Protocols identify all generator and evaluation seeds.

Implementation checks. Stage3 verifies 480 diagonal cells and 960 crossed finite-memory controls. If event/control observations agree from exit e, a fixed current context a, archive context b and window W give identical half-widths from $e + \max(a, b + W)$. This is an input-dependency check; persistent adaptive controller states need not obey it. It does not assert market recovery within 192 days. TCN64 full-sequence inference is checked against reset windows before test access. Canonical pulse horizons preserve identical entry prefixes across durations. Saved-output audits reproduce validation losses, sampling digests, stopping decisions and response features.

Component means and absolute interval quality. The first table uses Gaussian D128/A3 and all 15 fits per architecture, with route $C_{b64} + A_{a16}$; the second uses context64 recovery64 event paths. Higher coverage can accompany wider intervals and a worse score.

Model	Phase	Total	Current	Archive
GRU	Entry	-0.2071	-0.1302	-0.0769
GRU	Recovery64	-0.0326	+0.0504	-0.0830
TCN	Entry	-0.0085	+0.0611	-0.0697
TCN	Recovery64	-0.0396	+0.0210	-0.0606

Model	Interval	Score	Coverage	Width
GRU	Native	4.986	94.42%	4.544
GRU	W32	5.245	91.48%	4.439
GRU	W128	5.299	95.09%	4.902
TCN	Native	4.938	94.26%	4.470
TCN	W32	5.229	91.66%	4.452
TCN	W128	5.283	95.16%	4.887

Standard score geometry. For Gaussian $Y \sim N(0, \tau^2)$, expected interval score is

$$R(h; \tau) = 2h + \frac{4}{\alpha}\{\tau\phi(h/\tau) - h\,\overline{\Phi}(h/\tau)\}.$$

With $h = sq$ and $z = h/\tau$, the mixed derivative $R_h + hR_{hh} = 2 - (4/\alpha)\{\overline{\Phi}(z) - z\phi(z)\}$ changes sign. A fixed-archive scale-by-quantile contrast is its oriented rectangle integral. These calculus facts explain why a signed interaction need not rank memory quality. An earlier seed4101 Gaussian/t5 innovation-integration check preserves its interaction directions but says nothing decisive about initialization variability.

Validation probes and rank checks. All 30 models retain raw-input gradients and older-block permutations on 256 saved validation targets, plus 64 separate validation pulse paths. Panel-specific scatterplots are exploratory: no equivalence test, meta-regression or predictive model selection was performed. For rank sensitivity, all 92,160 original per-path components and 1,920 crossed cell means reproduce Stage3 within 1.8×10^{-15}; 960 candidate-rank negative controls pass. Randomization is integrated after each nonlinear metric, including positive-part oracle deficits. New common bootstrap draws support paired rule changes, so original-rule bootstrap endpoints differ slightly from the historical Stage3 table while point estimates and raw outputs remain unchanged. The 97.5% intervals adjust for two phases separately within each rule/change series, not jointly across rules. Secondary 95% intervals are unadjusted; all rank analyses remain post hoc.

B Financial extension and complete comparisons

Data and chronology. Each currency's percentage log return $100\log(P_t/P_{t-1})$ is divided, without demeaning, by its training standard deviation (ddof1). The archive contains informational reference prices, not point-in-time vintages or executable prices. Ten pooled neural fits use Appendix A's models and optimizer, uniform currency sampling and seeds 4301–4305 on local CPU. Both contexts select one common checkpoint by 2019 NLL. Classical scales recurse continuously across boundaries; calibrators have separate states for each currency and forecast stream.

ACI uses $\gamma = .02$ and window128 with clipped error rate. Author SF-OGD/SAOCP use $\sqrt{3}\max_{2019}|y|/s$ as the fixed scale heuristic, without clipping scores; 29 of 56 distinct streams exceed it during replay. SAOCP lifetime is 8. Observed 2020 outcomes update calibration but are excluded from reported metrics; 2021 begins without resets. Strong controllers use all ten context64 fits. Native/rolling/ACI retain both contexts, including all eight rolling crossed cells.

Complete context64 absolute-score table.

Model	Native	W32	W128	ACI128	SF-OGD	SAOCP
Constant	3.740	3.226	3.244	3.222	3.232	3.298
EWMA16	3.168	3.373	3.222	3.287	3.373	3.545
EWMA64	3.195	3.277	3.218	3.267	3.288	3.395
GARCH	3.240	3.288	3.192	3.249	3.292	3.427
GRU64	3.178	3.374	3.219	3.284	3.355	3.495
TCN64	3.166	3.342	3.204	3.261	3.319	3.450

Entries are means of per-date IS in per-currency training-standard- deviation units, averaging neural seeds before equal currency weights. All corresponding coverage, width, per-currency, annual and individual-fit results remain in the saved tables; 2026 is partial through 9 October. EWMA16's native mean IS (3.168) is close to the neural means. With SF-OGD and SAOCP, a constant base scale plus adaptive calibration has lower mean IS than the other listed combinations, with coverage 90.01% and 90.75%. These descriptive means do not establish a deployment winner. For GBP, both neural native mean scores are instead higher than GARCH; pooled ordering is not universal.

Dependence and uncertainty. Paired circular date-block bootstrap uses 2000 draws, seed 2026101802, block128 as the main analysis and 20/60/256 as sensitivity. The same date draws apply to every asset, fit, method and crossed cell. Models are fixed; these exploratory unadjusted 95% intervals do not measure retraining uncertainty or guarantee future performance under arbitrary distribution shift. The predeclared context64 contrasts are GRU-minus-GARCH, TCN-minus-GARCH and GRU-minus-TCN under each identical calibrator. The GRU-native/GARCH difference is -0.0623 with block128 interval $[-0.1212, -0.0007]$, but its block256 interval includes zero. TCN/GARCH rolling128 and ACI intervals include zero at all four block lengths.

Financial pathway contrast.

Model	J_{FX}	C_{b64}	A_{a16}
GRU	-0.0061	+0.0145	-0.0206
TCN	+0.0171	+0.0175	-0.0004
GRU$-$TCN	-0.0232	-0.0030	-0.0202

At block128, total 95% intervals are $[-0.0145, +0.0024]$ for GRU and $[-0.0008, +0.0366]$ for TCN; the paired architecture difference is -0.0232, interval $[-0.0408, -0.0059]$. These conditional exploratory intervals do not establish a training-population difference. Both routes sum to J_{FX}, with order-dependent components: TCN archive means are -0.0004 and $+0.0085$, both with intervals including zero. Unlike synthetic J, it contains no counterfactual subtraction, known volatility scale or labelled recovery. It cannot establish that a real market shock caused a pathway effect.